ONE-DAY APPLIED LEARNING PROGRAM

# INNOVATION IN ACADEMIA

A faculty transformation lab for stronger patents, funded research and industry relevance

**VALUE PROMISE: Research novelty • Patent potential • Fundable partnerships**

01
PROGRAM

| AUDIENCE | DURATION | FORMAT | CUSTOM PROGRAM |
|---|---|---|---|
| **Teachers & academic leaders** | **1 day \| 9:30-17:00** | **Hands-on studio** | **Discuss with us** |

## Why this matters now

FOR ACADEMIC LEADERS

**Turn faculty expertise into a visible pipeline of novel research, defensible IP and fundable partnerships.**

WHY STUDENTS NEED IT

Research scholars learn to identify consequential problems, build inventive confidence and see how research becomes IP and real-world impact.

WHY ACADEMICIANS NEED IT

Faculty move beyond incremental publication, strengthen novelty and create credible paths to patents, funded research and industry partnerships.

## What we will explore

- Build an inventive mindset for research, teaching and institutional challenges.
- Separate genuine novelty from routine extensions of existing work.
- Use structured frameworks to connect evidence, IP and stakeholder value.

## What participants gain

- **Find novelty:** identify innovative, relevant problems worth solving.
- **Escape the publication trap:** move from incremental papers to defensible contribution.
- **Create IP:** shape novel, non-obvious and patent-ready concepts.
- **Attract support:** build credible industry and government-funding pathways.

INSTITUTIONAL RETURN

**A stronger pipeline of patentable research, fundable proposals and industry-facing faculty initiatives.**

INNOVATION IN ACADEMIA

# From novelty to evidence, IP and partnership

A clear one-day journey, followed by reusable tools and a practical continuation path.

## One-day learning journey

| | | |
|---|---|---|
| 09:30-10:15 | **Beyond the next paper** | Recognise the iterative publication trap and define meaningful impact. |
| 10:15-11:15 | **Novel problem discovery** | Scan tensions, unmet needs, anomalies and emerging opportunity landscapes. |
| 11:30-12:45 | **Storytelling for relevance** | Connect evidence, beneficiaries, urgency and a compelling research question. |
| 13:45-15:00 | **Framework lab** | Use TRIZ contradictions, inventive principles and structured idea generation. |
| 15:15-16:15 | **Reverse brainstorming and IP lens** | Expose failure modes and test novelty and non-obviousness. |
| 16:15-17:00 | **Impact pathway** | Map experiments, patent steps, industry partners and suitable funding routes. |

## Who should attend

- University and college teachers
- Heads of department and deans
- Research supervisors and lab leaders
- Curriculum, accreditation and faculty-development teams

## Methods & tools

- Novel problem and opportunity landscapes
- Research storytelling for relevance
- TRIZ contradictions and principles
- Reverse brainstorming
- Publication-to-patent decision lens
- Industry collaboration and funding canvas

## Takeaway toolkit

- A portfolio of consequential, underexplored research problems
- A novelty and non-obviousness canvas for a selected idea
- Concepts generated using TRIZ and reverse brainstorming
- A 30-day experiment, IP and stakeholder-engagement roadmap

## Custom program

**Custom programs can be discussed**

The scope can be shaped around the cohort, institutional priorities and desired outcomes. Workbook, innovation tool cards, IP lens, funding canvas and participation certificate can be included.


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THE INNOVATION FRAMEWORK

# Three pillars of innovation

An idea becomes strategically interesting when it is **unique**, **hard to do** and **value adding** at the same time. The programs build student capability across all three.

01 UNIQUENESS

### Meaningfully different

Students learn to ask whether the problem, insight or solution is genuinely different from what already exists - not merely a new implementation of a familiar idea.

**SKILLS BUILT**

Problem discovery, reframing, prior-art awareness, assumption breaking, divergent thinking and TRIZ.

02 HARD TO DO

### Technically defensible

Innovation needs technical or execution depth. Students learn to work with constraints, contradictions, evidence and experiments so the idea is more than a superficial concept.

**SKILLS BUILT**

Systems thinking, technical trade-offs, experimentation, prototyping, non-obviousness and execution discipline.

03 VALUE ADDING

### Creates real value

A clever idea is not automatically valuable. Students learn to connect innovation to a real user, institution, industry or societal need and define the change it should create.

**SKILLS BUILT**

Stakeholder insight, outcome framing, evidence, adoption thinking, value articulation and storytelling.

## Why all three have to come together

| | |
|---|---|
| **Unique + valuable, but easy to copy** | Interesting, but difficult to defend or sustain. |
| **Unique + hard, but low value** | Technically impressive, but unlikely to matter outside the project. |
| **Hard + valuable, but not unique** | Useful engineering, but limited differentiation or innovation claim. |
| **Unique + hard + value adding** | **A stronger foundation for defensible innovation and meaningful impact.** |

## How I help students build the three muscles

**DISCOVER**
Find tensions, unmet needs and consequential problems.

**CHALLENGE**
Question assumptions and seek a genuinely different angle.

**BUILD**
Work through constraints, contradictions and technical depth.

**VALIDATE**
Test evidence, stakeholder value, utility and feasibility.

**COMMUNICATE**
Turn the work into a research, patent or industry story.

THE CORE HABIT

**Do not ask only, “Can we build it?” Ask: “Is it different? Is it difficult enough to be defensible? Does it create enough value to matter?”**



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FROM INNOVATION CAPABILITY TO OUTCOMES

# What the three pillars are designed to enable

The objective is not idea generation for its own sake. Students learn to convert differentiated, substantive and useful ideas into stronger research, invention and industry-facing evidence.

## Outcome 01 - Impactful research

**UNIQUENESS**
A consequential question, clearer research gap and a contribution that is meaningfully different.

**HARD TO DO**
Technical depth, rigorous methods, experiments, difficult trade-offs and credible evidence.

**VALUE ADDING**
A problem that matters to a field, stakeholder, industry, society or future system.

**STUDENT-OWNED EVIDENCE**
Problem landscape • novelty map • research question • experiment plan • contribution statement • publication pathway.

## Outcome 02 - Patents and defensible IP

**UNIQUENESS**
Prior-art awareness and a clearly differentiated mechanism, method or system element.

**HARD TO DO**
Non-obvious technical substance, constraints and implementation depth that make the idea defensible.

**VALUE ADDING**
Utility, adoption potential and a reason the invention should exist beyond novelty alone.

**STUDENT-OWNED EVIDENCE**
Invention canvas • prior-art search terms • novelty hypothesis • non-obviousness rationale • prototype or evidence plan.

## Outcome 03 - Industry-ready innovators

**UNIQUENESS**
Spot opportunities others miss and frame a differentiated response rather than repeating obvious solutions.

**HARD TO DO**
Move from idea to execution: handle constraints, collaborate, experiment, learn and improve.

**VALUE ADDING**
Connect technical work to business, user and societal outcomes and explain why it matters.

**STUDENT-OWNED EVIDENCE**
Problem brief • prototype or concept • decision rationale • outcome story • pitch • 30/60-day action plan.

**WHAT ACADEMIC LEADERS CAN REVIEW AFTER A COHORT**

Problem portfolio • research directions • possible IP candidates • experiment plans • industry-facing pitches • participant capability evidence • recommended next steps.

*The program builds capability and stronger pathways. Publication, patent and hiring outcomes are not guaranteed and depend on the quality of subsequent work, evidence, review and execution.*


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